

Exercise 1 – Address Decomposition

A system uses 32-bit virtual addresses and a page size of 4 KB.

Answer the following questions:

1. How many bits are used for the page offset?
2. How many bits are used for the Virtual Page Number (VPN)?
3. Decompose the virtual address 0x12345678 into VPN and offset.

Solution

A page size of 4 KB corresponds to:

4 KB = 4096 bytes = 2^{12} bytes

Therefore, 12 bits are required for the page offset.

Since the virtual address is 32 bits wide:

VPN bits = $32 - 12 = 20$ bits

To decompose 0x12345678:

Offset = lower 12 bits = 0x678

VPN = upper 20 bits = 0x12345

Final answer:

VPN = 0x12345

Offset = 0x678

Exercise 2 – Physical Address Translation

A system uses:

- 32-bit virtual addresses
- 4 KB pages
- the following page table mapping:

VPN 0x12345 -> PFN 0xABCDE

Translate the virtual address 0x12345678 into a physical address.

Solution

The offset is determined by the page size.

4 KB = 2^{12} bytes, therefore the offset occupies 12 bits.

Virtual address: 0x12345678

VPN = 0x12345

Offset = 0x678

The page table maps:

VPN 0x12345 -> PFN 0xABCDE

The physical address is obtained by concatenating PFN and offset.

Physical address = 0xABCDE678

Exercise 3 – Single-Level Page Table Size

Consider a system with:

- 48-bit virtual addresses
- page size = 4 KB
- each Page Table Entry (PTE) = 8 bytes

Compute:

1. Number of virtual pages
2. Total size of a single-level page table

Solution

Page size = 4 KB = 2^{12} bytes.

Number of virtual pages:

$$2^{48} / 2^{12} = 2^{36} \text{ pages}$$

Each PTE requires 8 bytes.

Total page table size:

$$2^{36} \times 8 \text{ bytes} = 2^{39} \text{ bytes}$$

$$2^{39} \text{ bytes} = 512 \text{ GB}$$

Exercise 4 – TLB Reach

A processor contains a TLB with 512 entries.

Compute the TLB reach for:

1. 4 KB pages
2. 2 MB pages

Solution

TLB reach is defined as:

$$\text{TLB Reach} = \text{Number of Entries} \times \text{Page Size}$$

Case 1: 4 KB pages

$$512 \times 4 \text{ KB} = 2048 \text{ KB} = 2 \text{ MB}$$

Case 2: 2 MB pages

$$512 \times 2 \text{ MB} = 1024 \text{ MB} = 1 \text{ GB}$$

Exercise 5 – Effective Memory Access Time (EMAT)

A processor has:

- TLB hit time = 1 cycle
- TLB miss rate = 2%
- Page walk penalty = 120 cycles

Solution

EMAT formula:

$$\text{EMAT} = \text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$$

Substituting the values:

$$\text{EMAT} = 1 + 0.02 \times 120$$

$$\text{EMAT} = 1 + 2.4$$

$$\text{EMAT} = 3.4 \text{ cycles}$$

Exercise 6 – TLB Miss Penalty

A system uses 4-level paging.

Assume:

- each memory access requires 100 cycles
- all page table entries are already in DRAM

Compute the latency of a TLB miss.

Solution

With 4-level paging, translation requires:

- 4 accesses for page table traversal
- 1 additional access for actual data

Total accesses = 5

Latency = 5×100 cycles = 500 cycles

Exercise 7 – FIFO Page Replacement

A system has 3 physical frames.

Reference string:

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Simulate FIFO page replacement and compute the number of page faults.

Solution

Frames initially empty.

1 -> fault
2 -> fault
3 -> fault
4 -> replace 1 -> fault
1 -> replace 2 -> fault
2 -> replace 3 -> fault
5 -> replace 4 -> fault
1 -> hit
2 -> hit
3 -> replace 1 -> fault
4 -> replace 2 -> fault
5 -> hit

Total page faults = 9

Exercise 8 – LRU Page Replacement

Using the same reference string from Exercise 9 and 3 frames, simulate LRU replacement.

Solution

Simulation:

1 -> fault
2 -> fault
3 -> fault
4 -> replace 1 -> fault
1 -> replace 2 -> fault
2 -> replace 3 -> fault
5 -> replace 4 -> fault
1 -> hit
2 -> hit
3 -> replace 5 -> fault
4 -> replace 1 -> fault
5 -> replace 2 -> fault

Total page faults = 10

Exercise 9 – Thrashing Analysis

A system has:

- physical memory = 8 GB
- application working set = 16 GB

Explain what is likely to happen.

Solution

The working set exceeds available physical memory.

The operating system continuously swaps pages in and out of memory.

This produces:

- high page fault rate
- heavy disk activity
- low CPU utilization

The system enters a condition called thrashing.

Performance collapses because the CPU spends more time waiting for paging operations than executing instructions.

Exercise 10 – Combined Cache and TLB Miss

A system has:

- TLB hit rate = 99%
- TLB miss penalty = 100 cycles
- L1 cache hit time = 1 cycle
- L1 miss rate = 5%

- L1 miss penalty = 40 cycles

Compute:

1. Effective translation overhead
2. Effective cache access time

Solution

Step 1 – Translation overhead

$$\text{EMAT} = 1 + 0.01 \times 100$$

$$\text{EMAT} = 2 \text{ cycles}$$

Step 2 – Cache AMAT

$$\text{AMAT} = \text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$$

$$\text{AMAT} = 1 + 0.05 \times 40$$

$$\text{AMAT} = 1 + 2$$

$$\text{AMAT} = 3 \text{ cycles}$$

Exercise 11 – Page Fault Impact on Performance

A page fault requires 8 ms to load a page from disk.

Assume the processor clock frequency is 2 GHz.

Estimate how many processor cycles are lost during a page fault.

Solution

$$\text{Clock frequency} = 2 \text{ GHz} = 2 \times 10^9 \text{ cycles/s.}$$

$$\text{Page fault latency} = 8 \text{ ms} = 8 \times 10^{-3} \text{ s.}$$

Lost cycles:

$$(2 \times 10^9) \times (8 \times 10^{-3})$$

$$= 16 \times 10^6 \text{ cycles}$$

$$= 16 \text{ million cycles.}$$